Final Report — Exposure-Adjusted Industrial Fire & Explosion Incident Analysis in Turkey

Motivation

Industrial fires and explosions carry consequences that extend well beyond the immediate incident: worker casualties, production losses, environmental contamination, and long-term liability. What makes them particularly frustrating from a policy standpoint is that most are preventable. This project grew out of a straightforward question "Do industrial incident patterns in Turkey follow any detectable structure in time and space, and does organized industrial zone presence actually explain where incidents concentrate?" Answering that question required building a dataset that did not exist in ready-to-use form, then being honest about what the numbers do and do not support.

Data Source

The backbone of the dataset comes from the annual industrial fire and explosion reports published by the TMMOB Kimya Mühendisleri Odası İstanbul Şubesi. For the 2017–2023 period, incident-level rows were pulled directly from the Ek/Ek-1 appendix tables. The 2024 edition presented a complication: its appendix was image-based, making automated OCR extraction unreliable. Once a manually verified Excel version of the 2024 table was available, it was converted and brought into the pipeline as data/raw/kmo2024_manual.xlsx. The combined cleaned dataset covers 3,852 incidents spanning 2017 through 2024.

To give the incident data geographic context, province-level organized industrial zone figures from OSBÜK were merged in — specifically OSB counts, operational OSB counts, total OSB area in hectares, and total parcel counts per province.

Weather data was originally planned through Open-Meteo, but rate limiting made a full national pull impractical. The NASA POWER Daily API was used instead, yielding a province-day panel of 216,228 rows across 74 provinces. Variables include mean, minimum, and maximum temperature, relative humidity, maximum wind speed, and precipitation. Matching on province and incident date, 3,430 of the 3,852 incidents were successfully linked to weather observations.

Data Preparation

Standardizing the raw inputs took considerable effort. Dates, event type labels, casualty counts, severity classifications, city and district name variants, OSB flags, sector codes, and calendar-derived fields all required normalization across eight years of source documents. For temporal modeling, additional features were generated: year, month, day of week, season, weekend and holiday indicators, cyclic sine/cosine encodings for month and day of week, and the matched daily weather covariates. Cells with missing or ambiguous source values were left as-is rather than imputed, since fabricating values in an incident registry introduces more problems than it solves.

Two validation figures worth noting: the 2021 appendix extraction yielded 394 rows — 358 fires, 36 explosions, and 78 Istanbul incidents. The 2024 table added 720 rows, of which 694 were fires and 26 explosions.

Analysis

Exploratory work produced visualizations covering monthly distributions, day-of-week patterns, geographic concentration, sector breakdown, severity composition, OSB exposure alignment, ignition source categories, and incident causes. All figures are saved in the figures/ directory.

The hypothesis tests produced the following:

Temporal calendar pattern: A Kruskal-Wallis test on monthly incident counts across 2017–2024 returned $H = 11.81$, $p = 0.378$, epsilon-squared = 0.010 — not significant at alpha 0.05. Descriptively, summer and autumn post higher totals than winter and spring, and 2024 shows peaks in November, July, and August, but these fluctuations do not hold up as a stable seasonal signal across the full eight-year window.

Day-of-week pattern: Weekdays consistently outpace Sunday. Monday leads with 601 incidents; Sunday sits at the bottom with 418. This aligns with the intuition that incident exposure tracks operational activity, though without production schedules or worker-hours data, the interpretation remains descriptive.

Environmental covariates: Spearman correlations between province-day incident frequency and weather variables are statistically detectable but practically small — mean temperature $\rho = 0.034$, maximum wind speed $\rho = 0.048$, humidity $\rho = 0.009$, precipitation $\rho = -0.011$. Extreme-heat days show a marginally higher mean incident count, but the effect size does not warrant strong conclusions.

Spatial/OSB concentration: Provinces with matched OSB exposure report significantly more incidents than those without — Mann-Whitney $p = 1.70\text{e-}29$.

Exposure correlation: Province-level incident counts track OSB parcel count closely, Spearman $\rho = 0.807$ ($p = 1.17\text{e-}17$), and OSB area similarly, $\rho = 0.764$ ($p = 6.02\text{e-}15$).

Severity associations: The sector–severity relationship is statistically significant, chi-square $p = 6.51\text{e-}11$. After the 2024 data was added, however, province-level OSB exposure no longer predicts severity in a statistically meaningful way, $p = 0.342$.

Machine Learning

Severity classification drew on temporal features (cyclic month and day-of-week encodings, weekend and holiday flags), spatial features (province identifiers, Istanbul flag, OSB exposure variables), sector, event type, and province-day weather where available. Three classifiers were evaluated: logistic regression, random forest, and XGBoost. Random Forest produced the best cross-validated macro-F1, approximately 0.496, with test accuracy around 0.905. The SHAP summary plot is stored at figures/14_shap_summary.png.

Findings

The clearest result in this dataset is geographic rather than temporal. Incident counts track industrial capacity — provinces hosting more OSB parcels and larger OSB footprints consistently report more fires and explosions. This should not be read as evidence that OSBs are inherently more hazardous; it reflects the obvious fact that more industrial activity produces more incidents in absolute terms. The value of the exposure-adjusted perspective is precisely that it separates provinces where high counts are simply a function of industrial scale from those where incident intensity per parcel is genuinely elevated.

The temporal side of the analysis matters mainly for what it rules out. Monthly seasonality is not statistically robust once the full eight-year dataset is assembled. Weather correlations exist but are weak. The defensible summary is: spatial and exposure structure is strong, sector–severity association is moderate, and environmental and calendar temporal effects are limited.

Limitations and Future Work

The TMMOB KMO reports are compiled from public sources and should not be treated as a complete incident registry — coverage gaps are likely. A substantial share of ignition and formation causes remain listed as unknown. OSBÜK exposure figures reflect current capacity rather than year-by-year historical values, which limits the precision of any longitudinal exposure adjustment. The weather data is pinned to province city centers, not facility coordinates, and roughly 11% of incidents could not be matched to a valid province at all due to ambiguous location fields in the source PDFs.

Future iterations of this analysis would benefit from facility-level geocoordinates, official fire department response records with response times and resource deployment, historical OSB capacity reconstructed year by year, district-level industrial employment figures, production volume or shift schedule proxies, satellite nighttime light radiance as an activity intensity measure, and meteorological observations matched to actual facility locations rather than provincial centroids.